

Trees to Seas

A Dynamic Physiological-Suitability Engine for North Carolina Coastal Fisheries

Proof-of-concept proposal and modeling roadmap

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1. Project summary

North Carolina's fisheries are among the most heavily managed in the United States, yet several stocks keep declining — a pattern that echoes the Great Lakes collapse. The root limitation is that management treats estuarine water as a **static backdrop** and regulates species one at a time within fixed salinity zones. But every commercially and ecologically important species has known temperature, salinity, and dissolved-oxygen tolerances, and estuarine conditions swing week to week. This project turns water into a **dynamic, predictive habitat layer**: drive per-species physiological **suitability envelopes** with high-frequency water-quality time series to map **when and where** each species is inside vs. outside its survivable range — resolving the sub-seasonal heat-and-hypoxia “squeeze” windows that a twice-a-year trawl survey structurally cannot capture.

Why now, and why this is fundable: the proof-of-concept does not rest on a theoretical argument. It is **validated against real biology**. The Duke Bass Connections oyster team has collected paired data — high-frequency environmental sensors (temperature, dissolved oxygen, salinity, pH, precipitation) *and* observed oyster mortality and growth — at working farm sites. That lets us test, directly, whether a mechanistic stress index predicts oyster survival. The engine is then generalized to the data-rich Neuse–Pamlico system.

Central hypothesis. For estuarine species, sub-seasonal variation in physiological habitat suitability (temperature × salinity × dissolved oxygen) is large, spatially structured, and largely invisible to a twice-yearly survey; a continuous suitability/stress index built from existing high-frequency data will reveal recurring summer hypoxia-and-heat squeeze windows that better explain mortality and habitat loss than static, salinity-zone, single-species management.

2. Background and motivation

2.1 Heavily managed, still declining

NC regulates each species individually, partitioned by static salinity zones, and leans on fishery-independent trawl surveys sampled only a few times per year. The result is biomass estimates built on temporally sparse snapshots, while the conditions that actually determine survival — summer heat, hypoxia, drought-driven salinity spikes, post-rainfall stratification — vary on the scale of days.

2.2 The Great Lakes cautionary parallel

The Great Lakes lost numerous fish taxa (26 extirpated from at least one lake, four globally extinct) under cumulative stressors plus species-by-species management. Lake sturgeon collapsed around the 1880s–1920s and never recovered. The lesson is not a single cause but the failure of a management paradigm that ignored cumulative, interacting stressors — exactly the blind spot a dynamic suitability model is built to address.

2.3 The “trees to seas” land-use drivers

NC's coastal plain is forest on a ~25-year harvest rotation, and the state is a top-three hog producer with dense poultry operations; clearing and runoff drive siltation, nutrient loading, and summer hypoxia that smother oyster reefs and silt over spawning beds. These drivers motivate

why suitability collapses occur where and when they do — they are the context, while the model focuses on the water-quality-to-physiology link.

2.4 The real, defensible gap (an important correction)

Do not pitch this as “the survey ignores the water.” NC DMF Program 195 (the Pamlico Sound trawl survey) is **stratified-random** and **does record** surface/bottom temperature, salinity, dissolved oxygen, secchi (turbidity), and precipitation at every grid. The defensible, verifiable gap is **coarse temporal resolution (June and September only) and under-use of those covariates** — not non-measurement. The whole proposal is built on that solid footing. (See Appendix A.)

3. Objectives

1. Lock the critique to the verified gap (twice-yearly snapshot resolution + covariate under-use), not the false “no covariates measured” claim.
2. Stand up a reproducible Python pipeline that ingests the oyster team's environmental + biological data with QC and daily resampling.
3. Compile a cited tolerance table (temperature, salinity, DO, pH min/optimum/max, plus degree-day base temperatures) for the focal species, flagging NC-specific vs. transferred values.
4. Implement per-species suitability envelopes $f(T,S,DO,pH) \rightarrow [0,1]$ combined into a geometric-mean Habitat Suitability Index, plus a separate degree-day phenology layer.
5. Validate the index against observed oyster mortality and growth (the headline result), and quantify the summer squeeze windows the snapshot survey misses.
6. Package everything as a runnable, journaled repository so additional species, sites, and the Neuse–Pamlico data plug in later.

4. Data — the key asset

4.1 Primary (in hand): the Duke Bass Connections oyster dataset

Source: github.com/oystersdukebc. Sites: CMAST (Morehead City), DURL (Beaufort), and — in 2025–26 — Stump Sound and Ward Creek.

Data	Variables	Role in the model
AverageCMASTtemp.csv / AverageDURLtemp.csv	Water temperature, dissolved oxygen (hourly), by site	Environmental drivers
environmental_data_{daily,hourly,5minute}.csv	Long-format: temperature, salinity, pH, precipitation	Environmental drivers (multi-resolution)
2025–26 XLSX (DO, pH, Salinity, Precip; Cleaned/ per site)	DO, Temp, pH, salinity (conductivity-derived)	Environmental drivers (4 sites)
MortalityContinuous.csv / OysterMortalityData_Processed.csv	Per-bag survivorship over a biweekly census, by strain & site	Response variable (validation target)

Data	Variables	Role in the model
MortalityBySiteBinned.csv	Survivorship by site/date with N, sd, se, ci	Response variable (aggregated)
growth_rates.csv	Growth rate by strain/treatment/site, with CIs and p-values	Response variable (growth)

This pairing of drivers and biological response is what elevates the proof-of-concept from a plausibility demo to a genuine validation.

4.2 Generalization targets (Neuse–Pamlico)

Source	Variables	Access
ModMon (Neuse River Estuary, UNC Paerl Lab)	Temp, salinity, DO, turbidity, chlorophyll (1994–2021)	SECOORA ERDDAP (erddapy), QARTOD flags
USGS Fort Barnwell (02091814)	River discharge (freshwater forcing)	USGS NWIS (dataretrieval)
NOAA CO-OPS Beaufort (8656483)	Water temperature (no salinity sensor)	CO-OPS API (requests)

Scope note for Dr. Rittschof. We chose “Neuse–Pamlico first,” but the richest *paired* data (environment + biological response) is the oyster-farm sites. The strongest path is to build and **validate** the engine where we have ground-truth biology (oyster sites), then extend the environmental engine to the Neuse–Pamlico. This is a deliberate refinement to confirm.

5. Approach and methods

5.1 Suitability envelopes and the Habitat Suitability Index

Each environmental factor maps to a [0,1] suitability via a transparent trapezoidal curve defined by min / optimum-low / optimum-high / max thresholds. Factors are combined into a single Habitat Suitability Index (HSI) using the geometric mean, so that any one lethal factor (for example, DO → 0) drives the whole HSI to zero. This is what captures a true hypoxia/heat “squeeze”; an arithmetic mean would mask it. A Liebig minimum-factor combiner is also provided and compared. The model is deliberately mechanistic and interpretable — not a machine-learned species-distribution model — because interpretability, not raw fit, is the point.

5.2 Degree-days (phenology layer, kept honest)

Degree-days predict the timing of development, spawning, and activity for cold-blooded animals and are well supported for pre-maturation growth (Neuheimer & Taggart 2007 found accumulated degree-days explained >92% of length-at-day variance across nine fish species). They are kept strictly separate from the survival envelope, because a plain degree-day sum breaks down near thermal extremes — upper-lethal limits and hypoxia are handled by the envelope, while degree-days only predict when favorable windows open.

5.3 Validation against observed biology (the payoff)

For each census interval at a site we accumulate a stress load from the suitability time series and test whether it rank-correlates with observed oyster mortality. As reported in §5.5, this oyster-mortality test came back null — the tolerant oyster stayed within its adequate range at these well-flushed sites — so the load-bearing validation is the Program 195 spatial-occupancy result below, not oyster mortality. The continuous suitability series is also contrasted with documented low-DO periods (§5.6).

5.4 Tooling: Python

Recommended: Python. The machine-readable route to the generalization data is SECOORA ERDDAP, and erddapy gives clean, scriptable queries that preserve QC flags; pandas/xarray, dataretrieval (USGS), matplotlib, and scipy cover the rest of the pipeline in one coherent stack that also matches the oyster team's GitHub workflow. R remains fully capable (the team already uses tidyverse/Shiny), and the repository's loaders read their R-exported CSV/XLSX directly.

5.5 Results to date — what is already validated (and what is not)

Headline result — independent, out-of-sample, and robust. Using each tow's own bottom temperature, salinity, and dissolved oxygen with the *literature-calibrated* envelopes (never fit to catch data), the Habitat Suitability Index positively and very significantly predicts where all three trawl species are caught across 4,312 NC Program 195 tows (1987–2021): Atlantic croaker $\rho = +0.20$, blue crab $+0.18$, southern flounder $+0.14$ (all $p < 1e-19$; presence-AUC 0.55–0.59). The signal is **robust to envelope uncertainty**: under $\pm 15\%$ perturbation of every threshold (300 Monte-Carlo draws) the correlation stays positive in 100% of draws for all three species — so it does not depend on the exact (often transferred) numbers.

Species-specific honesty. Controlling for year, the signal is robust for southern flounder (within-year median $\rho = 0.15$, positive in 89% of years), moderate and autumn-concentrated for croaker (0.11), and weak for blue crab (0.08, ~58% of years). The blue-crab result is **expected, not a defect**: 97% of tows fall inside the crab's salinity tolerance plateau and 82% inside its oxygen plateau, so those axes are mathematically flat across the well-mixed sound, and adult blue crab is a euryhaline, mobile, sex-segregated generalist that actively avoids hypoxic bottom water and buries below $\sim 10^\circ\text{C}$ — its realized location is not set by a static bottom reading. We keep the crab envelope as a labeled physiological-tolerance layer and flag a sex-/life-stage-structured model as the correct future path. Flounder, pinned to low-salinity fine sediment, behaves exactly as the method intends.

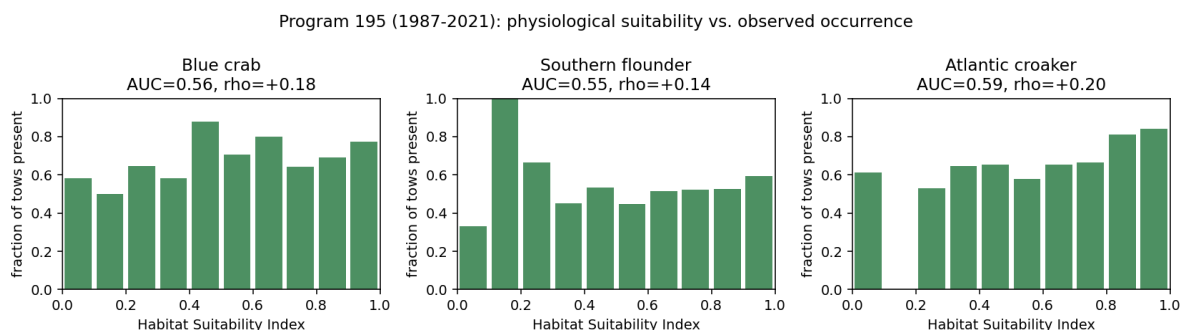


Figure 1. Independent validation: fraction of Program 195 tows where each species was caught, by habitat-suitability decile (suitability computed from each tow's own bottom T/S/DO using the

literature-calibrated envelopes). Occurrence rises with modeled suitability for all three species. Data: SEAMAP-SA Pamlico Sound Survey (NC DMF), used under the SEAMAP-SA Intellectual Property Protocol — not for redistribution.

5.6 The summer squeeze the survey can't see

The model's central claim — that sub-seasonal, bottom-water stress is large and survey-blind — is borne out directly by depth-resolved ModMon profiles (1994–2021). In the Neuse River estuary, **bottom water is hypoxic (DO < 2 mg/L) in 44% of warm-season profiles versus 0.5% at the surface**; it peaks in August (54% of bottom profiles; median bottom DO 1.5 mg/L; modeled croaker bottom-habitat suitability → 0) and is worst in July–August, between the survey's June and September visits (51% vs 37%). The surface looks pristine while the bottom — where demersal species live — suffocates. This is measured dissolved oxygen, in the survey's own system, with no model assumptions.

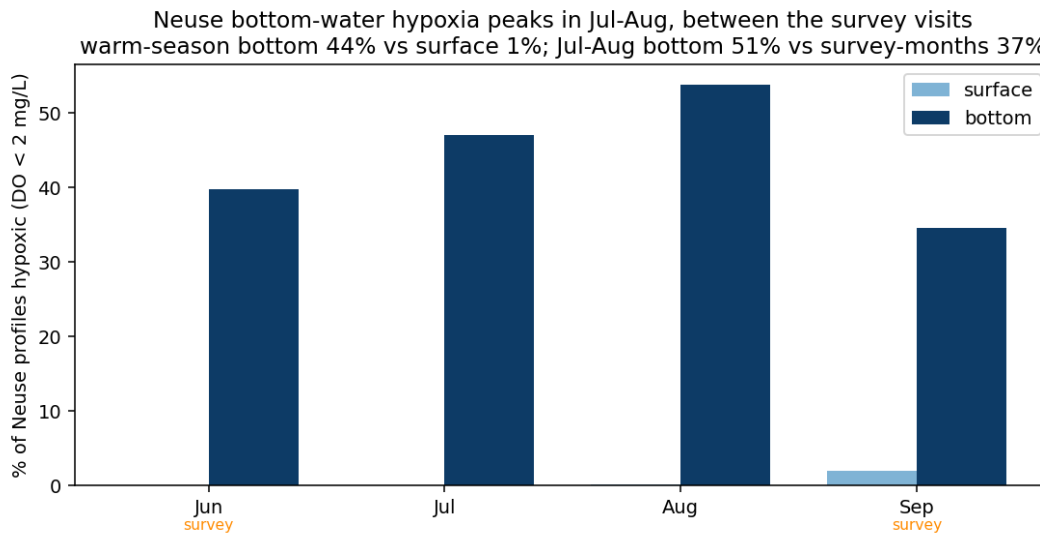


Figure 2. Neuse bottom- vs surface-water hypoxia by month (depth-resolved ModMon, 1994–2021). Bottom-water hypoxia peaks in July–August — between the June and September survey windows. Data: UNC ModMon via SECOORA ERDDAP.

Honest scope — what the model does NOT do. Two stringent tests returned null, and they sharpen the tool's purpose. (a) Annual abundance is **not** predicted by that year's mean habitat suitability (no positive correlation across 28 years) — year-to-year abundance is dominated by recruitment and fishing, not adult habitat. (b) Oyster mortality at the well-flushed CMAST/DUML farm sites is not explained by temperature/oxygen suitability; in fact the more oxygen-stressed site had *higher* survival (0.96 vs 0.80), because the tolerant oyster stayed within its adequate range and mortality there is governed by other factors (salinity, disease, handling). The conclusion: this is a **spatial and sub-seasonal habitat-stress lens** — validated for where mobile species occur and for resolving the bottom-water squeeze — not a stock-abundance or oyster-mortality predictor. Stating that scope is itself a contribution: it tells managers exactly what the tool is and is not for.

6. Research workflow — modeled on The AI Scientist

The project's **process** (not its AI content) is modeled on The AI Scientist (Lu et al., Nature 2026), which runs research as a single automated driver advancing through discrete stages, where each experiment is a journaled “node” that is automatically reviewed before the best one is carried forward. We adopt that scaffold because it gives a small project reproducibility, auditability, and explicit go/no-go gates.

The AI Scientist (Lu et al. 2026)	Trees to Seas implementation
Four-stage experiment progress manager	Stage 1 viability → Stage 2 envelope tuning → Stage 3 execution → Stage 4 ablation
Stage transitions on pre-defined criteria	e.g. Stage 2 advances only if the full HSI beats the temperature-only baseline
Each experiment is a journaled, typed node	Structured journal.jsonl + final_info.json + best_node.json + notes.txt
LLM judge selects the best node to seed the next stage	Driver selects the best node (review score) and carries the best combiner forward
Automated Reviewer (LLM ensemble, NeurIPS rubric)	Deterministic 5-dimension review rubric (upgradeable to an LLM reviewer)
Buggy nodes recorded, not fatal	Every stage is wrapped so failures are journaled, not crashes

Full mapping and rationale are in docs/methodology.md in the repository.

7. Eight-week roadmap

Because the primary data is already in hand, the schedule front-loads validation rather than data hunting.

Week	Goal	Deliverable
1	Lock scope + corrected framing; initialize repo; request FerryMon/post-2021 ModMon; sync data format with the oyster team	One-page scope + corrected-facts sheet; reproducible repo (done in v0.1)
2	Reliable ingestion + QC of all oyster sites/years (CSV + XLSX)	Clean dataset + data-coverage figure
3	Parameterize defensible, cited tolerance envelopes (NC vs. transferred)	Versioned tolerance table + factor-curve panel
4	Implement + unit-test the HSI engine; calibrate against mortality	Tested HSI module + per-site HSI time series
5	Add the degree-day phenology layer with honest caveats	GDD curves + predicted spawning/activity windows
6	Headline figures: when/where suitability-vs-stress; quantify the snapshot gap	Dynamic suitability maps + “what the snapshots miss” figure
7	Generalize: wire ERDDAP/USGS/NOAA; reproducibility hardening	One-command-reproducible repo; optional Neuse demo

Week	Goal	Deliverable
8	Write proposal narrative around figures; advisor review packet	Final proposal + roadmap + repo v0.2 + advisor question list

8. Deliverables

- This proposal + modeling roadmap document.
- A reproducible Python repository (treestoseas v0.1): ingestion → QC → HSI → degree-days → validation → figures, journaled per run.
- A cited species tolerance/parameter table with NC-specific vs. transferred values flagged.
- A tested HSI module (geometric-mean envelopes) and a separate degree-day module.
- The headline figure set: dynamic when/where suitability-vs-stress maps with observed oyster mortality overlaid.
- A corrected-facts crib sheet (Appendix A).
- An advisor packet: open questions and a data-coordination note for the oyster team; a generalization/validation plan for the Neuse–Pamlico.

9. Risks and mitigations

Risk	Mitigation
Pitch resting on the false “no covariates” premise	Already corrected: lead with temporal-resolution + under-use; cite SEAMAP-SA (Appendix A)
Tolerance thresholds transferred from other systems → false precision	Tag every value NC-specific vs. transferred; run a sensitivity pass; present HSI as a relative index
FerryMon access is email-only and routes have been down since 2019	Make the oyster sites + ModMon-via-ERDDAP primary; FerryMon a stretch goal
Degree-days over-promise near thermal extremes	Keep degree-days as a timing layer only; the envelope handles upper-lethal limits and hypoxia
Sensor gaps / cadence mismatch across sources	Resample to common daily/weekly cadence, carry QC flags, show coverage plots, never silently interpolate
Landings declines mistaken for abundance declines	State the distinction explicitly; use croaker suitability-vs-landings as a feature, not a hidden assumption
Scope creep into a validated SDM or multi-region build	Hard-commit to oyster sites, the focal species, a mechanistic model, and sufficiency-style validation

10. Open questions

For Dr. Rittschof

- Is the reframed critique (sub-seasonal resolution + covariate under-use, not “no covariates”) strong enough to anchor the proposal?

- Confirm the scope refinement: validate on the oyster sites first, then generalize the engine to the Neuse–Pamlico?
- Is the focal species set right (eastern oyster anchor; blue crab, southern flounder, Atlantic croaker)? Add river herring?
- Is the geometric-mean HSI (any lethal factor zeros suitability) the right combiner, or do you prefer a Liebig minimum-factor approach?
- What validation bar do you expect — is the sufficiency demo enough, or do you want a quantitative comparison against Program 195 catch indices?
- Should the land-use “trees to seas” driver narrative be carried explicitly, or kept to the significance section?

For the oyster team (Ty / Juliet / Mihir)

- What are the units, sensor models, and known QC issues for the CMAST/DUML/Stump Sound/Ward Creek sensors?
- What do the oyster strains (BS, CN, CS, SJ) and treatments (50/100 starting density) denote, and which are most comparable?
- Is salinity in the 2024–25 data conductivity-derived, and what conversion was used?
- Can we get post-2024 cleaned files and a stable data-format contract so the pipeline ingests new seasons automatically?

Appendix A — Corrected-facts crib sheet

The original brainstorming notes contained several factual errors, fact-checked against NOAA, USGS, NC DMF, UNC/Duke, and APNEP sources. Several are load-bearing.

Claim in the notes	Corrected
Menhaden killed in Chicago, 1964	Alewife (<i>Alosa pseudoharengus</i>) die-off, Lake Michigan / Chicago shoreline, 1967. Menhaden are absent from the Great Lakes.
~20 species extinct in the Great Lakes	26 fish extirpated from ≥ 1 lake, 4 globally extinct (DFO checklist). “~20” only as a loose figure.
Sturgeon collapsed in the 1960s–70s	Lake sturgeon collapsed ~1880s–1920s; ~1% of historic abundance today. The 1960s–70s is when blue pike went extinct.
Cisco/whitefish lost mainly to logging siltation	Primary drivers: overfishing + invasives; siltation contributing, not the lead cause, and not attributed to logging.
2nd-largest estuary in the world	“2nd-largest estuarine complex in the lower 48” (APNEP). Pamlico Sound is the largest embayed estuary in the world — keep distinct.
Program 195 ignores salinity/rainfall/turbidity/temperature	FALSE. Program 195 is stratified-random and records temp, salinity, DO, secchi (turbidity), precipitation at every grid. Real critique: twice-yearly resolution + covariate under-use.
NC #2 in hog production	#3 in total hog inventory (Iowa, Minnesota ahead); ~#2 in the breeding herd. State metric + year.
Hogs = #1 nutrient-pollution source in NC	Poultry has overtaken swine (~3× the N, ~6× the P; NC DEQ 2017).
Croaker/spot “crashed ~97%”	That is a landings decline, not abundance — NC surveys show croaker/spot remain among the most abundant estuarine fish.
Drought pushes crabs into freshwater and they die	Backwards: drought raises salinity; higher salinity correlates with lower crab abundance and faster <i>Hematodinium</i> spread.
FerryMon = clean continuous 20-yr record	FerryMon (not “Fairymon”) is NOT continuous: funding gap after ~2016; routes down since 2019/2021. Treat as a stretch goal.
Cape Hatteras NERR (Beaufort)	No Cape Hatteras NERR. NC NERR: Currituck Banks, Rachel Carson (Beaufort), Masonboro Island, Zeke's Island.

References

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